Radijator INŽENJERING

INDUSTRIAL HEATING AND ENERGY SOLUTIONS

COMPLETE PRODUCT CATALOGUE / 2026

INDUSTRIAL BOILERS FOR BIOMASS

Reliable high-output systems designed for efficiency, long service life and complete control of the combustion process.

01 ABOUT US

In business terms, Radijator Inženjering d.o.o. is the legal successor to the Radijator craft business, founded in 1991, whose main activity was the installation and maintenance of central heating systems. We manufactured our first hot water boiler for solid fuel in 1985.

The company has operated in its current form since 2002 and has made substantial progress year after year, always striving to lead in the adoption of new technologies, product quality and entry into new European markets.

As we expanded and improved production, we reached the point where we manufacture boilers using the most advanced technologies available worldwide. Our sheet metal cutting processes include laser cutting, CNC plasma cutting and CNC punching. We carry out welding using robots and automatic welding equipment. The annual growth in production is one of the clearest indicators of the quality of our products and services.

Today, Radijator Inženjering employs more than 350 people, including 40 graduate mechanical engineers who work every day to improve product quality.

ISO 9001:2008 quality management system certification confirms the consistent quality of both our products and our business operations.

Radijator Inženjering is a Serbian manufacturer of biomass boilers with a long tradition, recognised for reliable and technologically advanced heating solutions. By developing our own production processes and applying modern technologies, we create products that meet the highest market requirements for quality, efficiency and long service life.

Our production range includes boilers with outputs from 6 to 600 kW, intended for heating family homes, residential and commercial buildings, public institutions and industrial facilities. Our extensive product range and ability to build cascade systems allow us to offer an optimum solution for every building and every requirement.

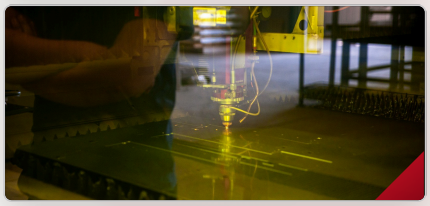
The quality of our products comes from carefully selected materials, precision manufacturing and rigorous quality control at every stage of the process. We develop every boiler to deliver maximum energy efficiency, reliable operation and long-term use with minimal maintenance costs.

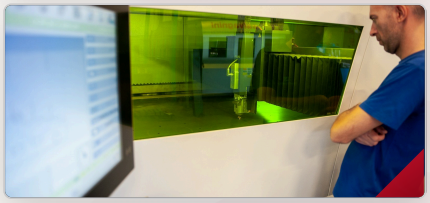
We pay particular attention to innovation and improvements in production technology to provide customers with modern solutions that combine a high level of automation, simple operation and maximum energy utilisation.

We design and manufacture all our products in accordance with applicable European standards and regulations for boilers, confirming their safety, reliability and high quality.

By choosing Radijator Inženjering boilers, you choose a Serbian product, proven quality and a partner whose experience, expert support and complete system solutions provide dependable support throughout the life of your heating system.

A company with 35 years of experience in designing, manufacturing and innovating boilers that heat thousands of buildings across Europe!





PRODUCTION

TECHNOLOGY AND QUALITY

PRODUCTION TO MODERN EUROPEAN STANDARDS

As production expanded and improved, we began manufacturing boilers using the latest technologies: laser cutting, CNC plasma cutting, CNC punching, robotic welding and automatic welding equipment.

Today, Radijator Inzenjering employs more than 350 people, including 40 graduate mechanical engineers who work every day to improve product quality.

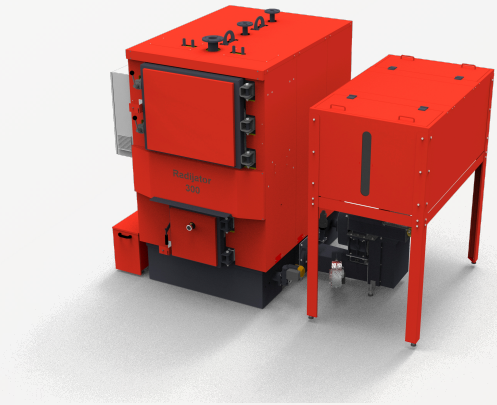
350+
employees

40
graduate mechanical engineers

EU
exports to 27+ EU countries

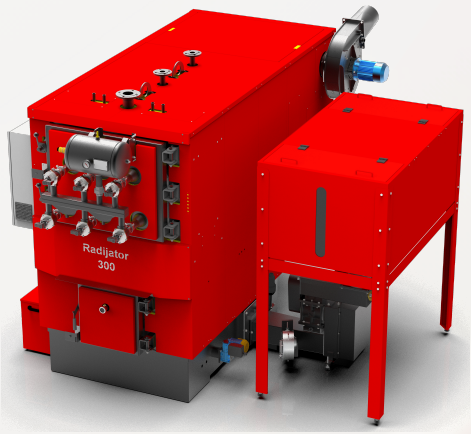
TECHNOLOGY / QUALITY / MARKET

02 PRODUCT RANGE - INDUSTRY



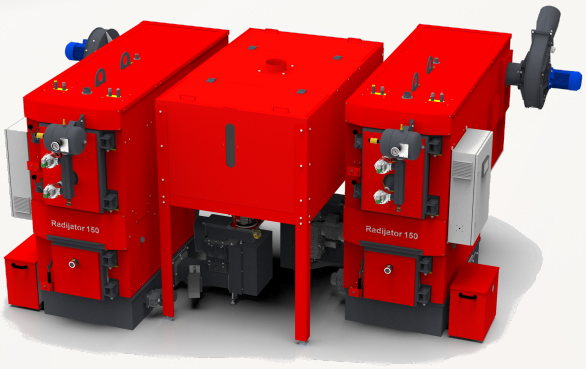
TKAN SERIES

Our industrial biomass boilers use boiler steel plates at least 6 mm thick. The heat exchanger uses seamless boiler tubes. Efficiency exceeds 90% when burning pellets. At higher operating loads, the flue gas temperature at the outlet ranges from 170 to 190 degrees Celsius and can always be checked on the control display. Available outputs range from 60 to 300 kW.



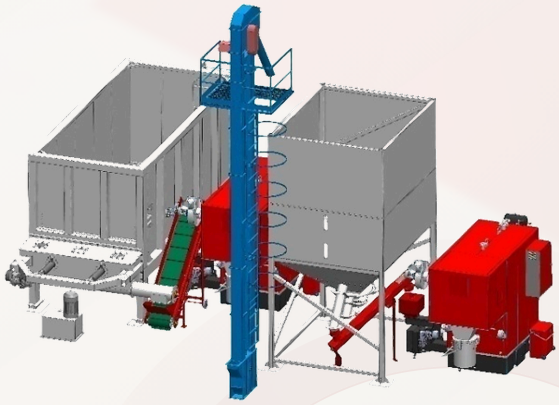
TKAN INTEGRA SERIES

This industrial biomass boiler is an improved version of the standard TKAN boiler. It features a refractory-lined combustion chamber, more advanced controls and a wider range of auxiliary equipment, providing greater reliability, improved combustion efficiency and broader applications. It uses boiler steel plates at least 6 mm thick and a heat exchanger made of seamless boiler tubes. Efficiency exceeds 90%. Available outputs range from 80 to 600 kW.



CASCADE SYSTEMS

Cascade systems combine two or more boilers in a single system with a shared pellet storage silo. This configuration provides greater total installed output, more reliable operation, balanced load distribution and higher system energy efficiency.

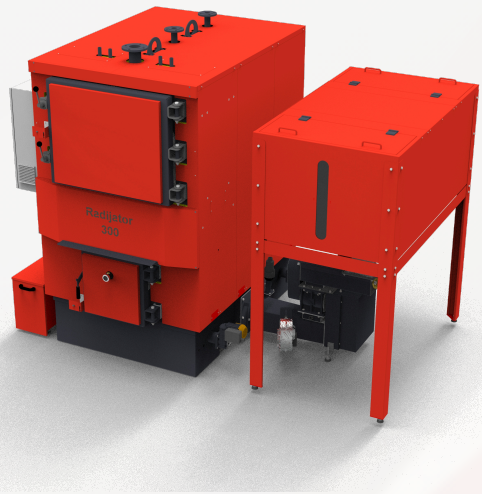


ADDITIONAL EQUIPMENT

Alongside boilers, we offer a complete range of auxiliary equipment for a fully functional heating system. The range includes pellet storage silos, screw conveyors, elevators, buffer tanks, controls and other equipment needed for reliable pellet transport, dosing and control, as well as safe and efficient operation of the entire system.

03 TKAN HOT WATER PELLET BOILER WITH AUTOMATIC FUEL FEED

(Available outputs: 60-300 kW.) RADIJATOR INŽENJERING developed the TKAN boiler to offer the market a boiler with mechanical and thermal properties specifically suited to biomass fuel. At the same time, the market consistently demands greater fuel flexibility, so the TKAN can also burn wood, which requires manual loading.



In its external design, combustion chamber dimensions, and fuel-loading and cleaning openings, the TKAN retains all the strengths of earlier models for which RADIJATOR INŽENJERING is known in the market. The boiler's water section and the transfer of heat from flue gases to water through a tubular heat exchanger are adapted to biomass. The use of a fan, providing forced draught, creates a longer flue gas path than in standard boilers. This also allows the use of flue gas flow devices known as turbulators, which further increase boiler efficiency. The turbulators are spirals made from a special material.

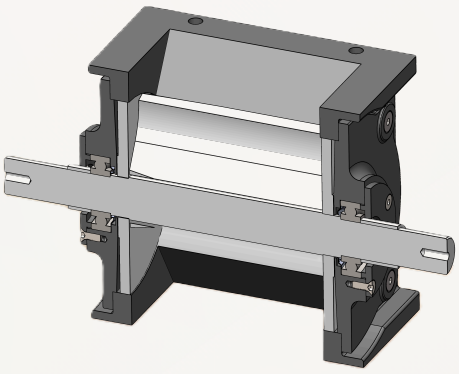
Efficiency exceeds 90% when burning pellets. Under normal operating conditions, the flue gas temperature at the outlet is approximately 160 °C, and at maximum output it remains below 180 °C. These values can be read on the display at any time.

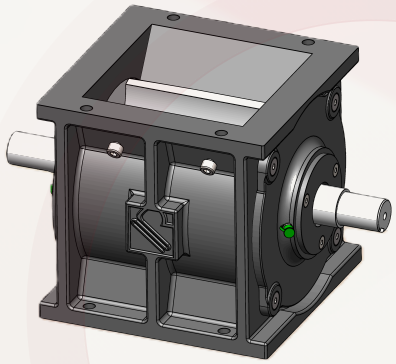
All components of the boiler's water section use seamless ST 35.4 tubes and boiler steel plates at least 5 mm thick, depending on boiler output. The plates are grade 1.0425 under the EU standard, or P265GH under the EUII standard. The combustion chamber operates on an underfeed principle: fuel moves vertically upwards from the conveying zone into the combustion zone. It uses substantial insulating materials and grey cast iron. Screw conveyors transport the fuel.

Table 1. Available outputs

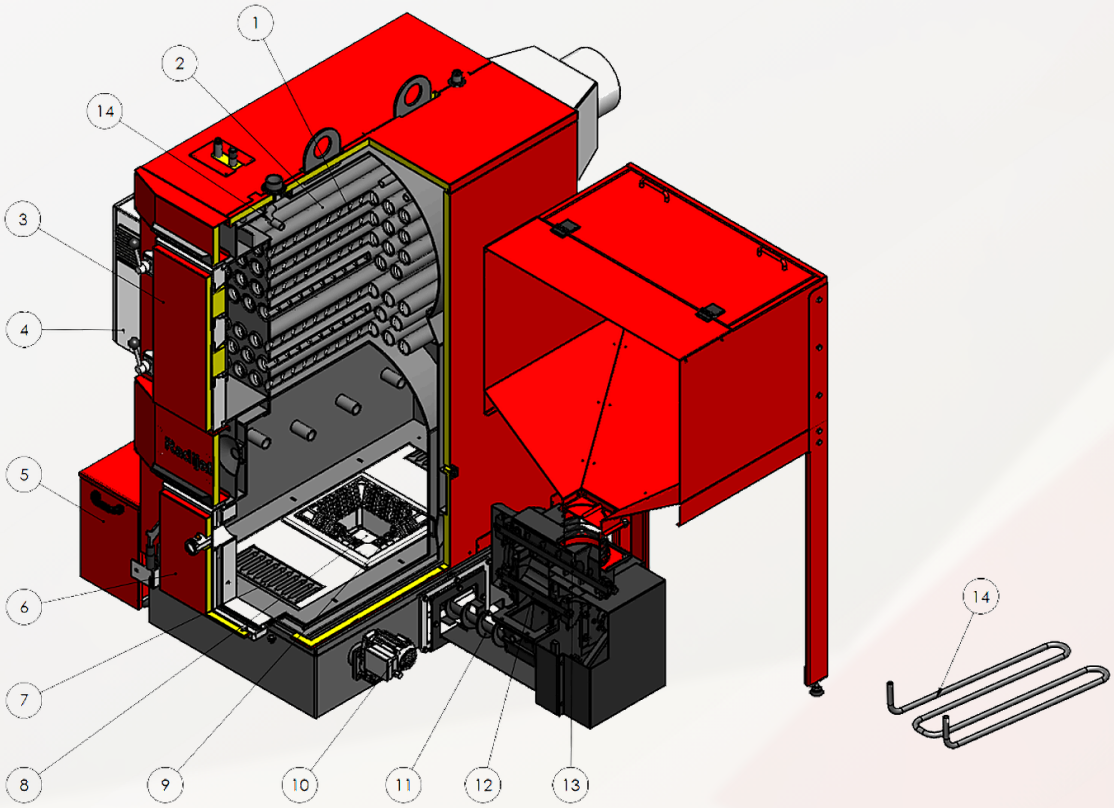
TKAN / TECHNICAL SPECIFICATIONS

BOILER TYPE	BOILER TYPE	BOILER TYPE	TKAN 60	TKAN 80	TKAN 100	TKAN 150	TKAN 200/250/300
		Unit					
Output	Output	kW	60	80	100	150	200/250/300
Operating pressure	Operating pressure	kPa	300	300	300	300	300
Test pressure	Test pressure	kPa	450	450	450	450	450
Boiler water volume	Boiler water volume	L (approx.)	276	368	460	690	920/1150/1380
Boiler weight	Boiler weight	kg	655	915	1073	1665	2260/2800/3080
Silo weight	Silo weight	kg	106	100	110	162	180/220/220
	As	mm	610	610	810	1010	1394
Dimensions	B	mm	1020	1020	1176	1456	1830
Dimensions	Hs	mm	1740	1736	1611	1875	1822/1822/2070
	V	L		290	305	680	1600/2330/3115
Pellet capacity of the silo	Pellet capacity of the silo	kg		180	190	410	1000/1500/2000



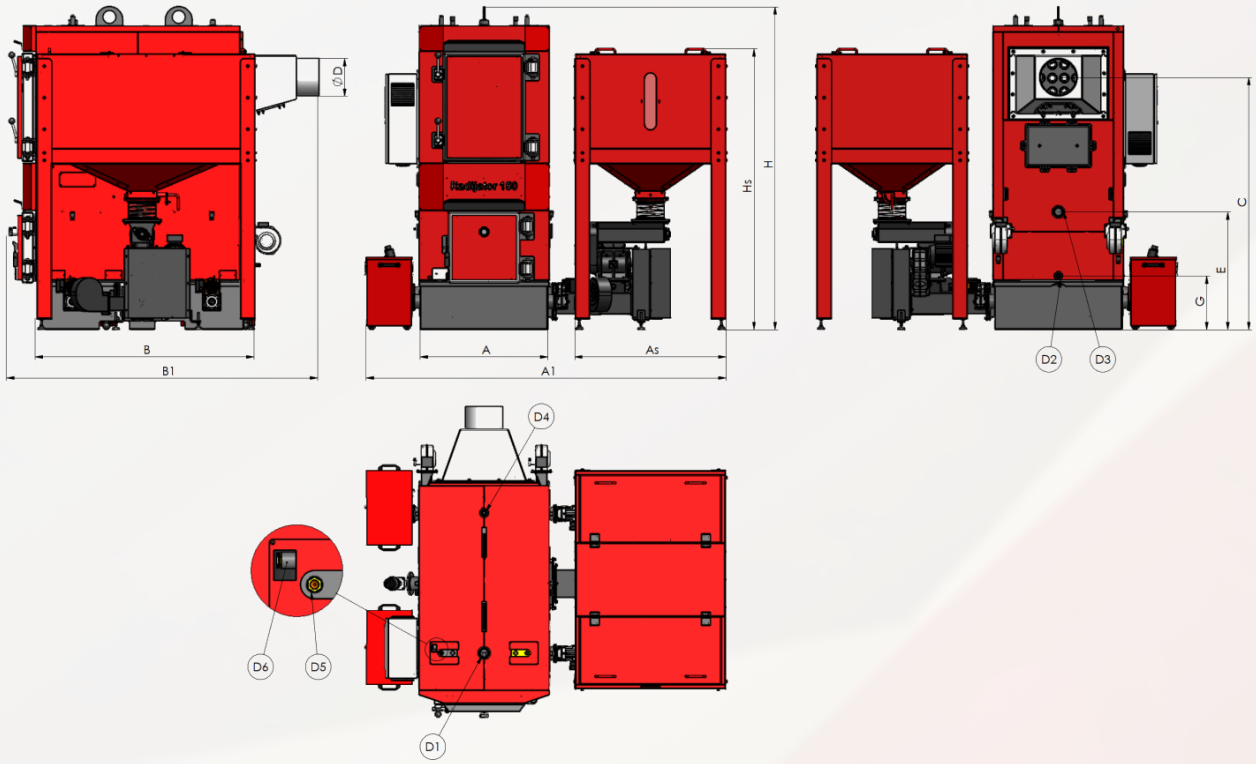


04 TKAN BOILER CROSS-SECTION



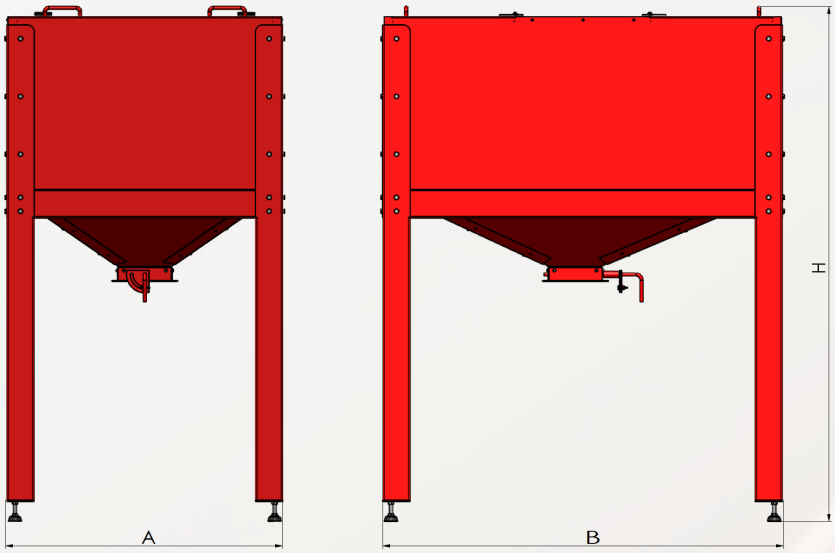
01 Turbulators	02 Tubular heat exchanger
03 Access door for cleaning the tubular heat exchanger and boiler	04 Electrical cabinet with controls
05 Ash container	06 Fuel-loading and ignition door
07 Auger for automatic ash removal from the combustion chamber	08 Boiler combustion chamber
09 Cast iron segments	10 Drive motor for the auger that automatically cleans the combustion chamber
11 Lower screw conveyor shaft	12 Rotary feeder (rotary valve)
13 Upper screw conveyor shaft	14 Thermal safety heat exchanger

05 TKAN BOILER DIMENSIONS



DIMENSIONS	DIMENSIONS	DIMENSIONS	TKAN 60	TKAN 80	TKAN 100	TKAN 150	TKAN 200	TKAN 250	TKAN 300
A	mm	mm	680	730	730	850	1005	1260	1260
A1	mm	mm	1355	1788	2080	2395	2945	3456	3455
As	mm	mm	610	610	810	1010	1394	1394	1394
B	mm	mm	1020	1020	1176 - 1286	1456	1830	1830	1830
B1	mm	mm	1518	1565	1727	2070	2121	2546	2546
C	mm	mm	1125	1573	1573	1681	2044	2114	2114
ØD	mm	mm	200	200	200	250	250	300	300
E	mm	mm	675	417	417	785	675	707	707
G	mm	mm	360	389	389	360	465	467	467
H	mm	mm	1564	1960	1960	2151	2547	2663	2663
Hs	in.	in.	1740	1736	1611	1875	1822	1822	2070
D1	in.	in.	6/4"	2"	2"	2"	DN80 NP6	DN80 NP6	DN80 NP6
D2	in.	in.	6/4"	2"	2"	2"	DN80 NP6	DN80 NP6	DN80 NP6
D3	in.	in.	1/2"	1/2"	1/2"	1/2"	1"	1"	1"
D4	in.	in.	3/4"	3/4"	3/4"	3/4"	DN40 NP16	DN40 NP16	DN40 NP16
D5	in.	in.	1/2"	1/2"	1/2"	1/2"	1/2"	1/2"	1/2"
D6	in.	in.	1/2"	1/2"	1/2"	1/2"	1/2"	1/2"	1/2"

06 TKAN SILO DIMENSIONS

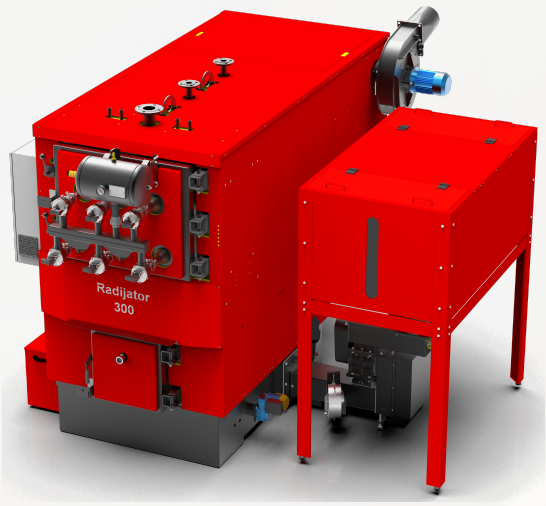


	DIMENSIONS	DIMENSIONS	DIMENSIONS	DIMENSIONS	DIMENSIONS
	A	B	H	V	Pellet capacity of the silo
	mm	mm	mm	L	kg
TKAN 80	606	1020	1736	290	180
TKAN 100	810	1176	1611	305	190
TKAN 150	1010	1456	1875	680	410
TKAN 200/250/300	1394	1830	1828	1600	1000
TKAN 200/250/300	1394	1830	2138	2330	1500
TKAN 200/250/300	1394	1830	2448	3115	2000

07 TKAN INTEGRA HOT WATER PELLET BOILER WITH AUTOMATIC FUEL FEED, DUST REMOVAL AND CYCLONE

(Available outputs: 80-600 kW.)

TKAN INTEGRA is a new generation of industrial biomass boilers, developed as an improvement on the standard TKAN model. It responds to market demand for greater automation, higher energy efficiency and more reliable operation, while retaining all the proven design solutions for which RADIJATOR INŽENJERING is known.

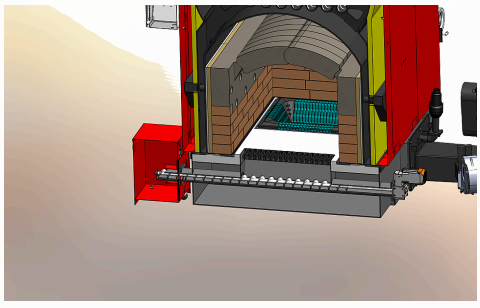


With its refractory-lined combustion chamber, improved combustion system, modern controls and more extensive standard equipment, TKAN INTEGRA provides stable operation, maximum energy utilisation and a long service life for all key components. The boiler structure uses high-quality boiler steel plates at least 6 mm thick, and the tubular heat exchanger uses seamless boiler tubes, ensuring high mechanical strength, reliability and a long system service life.

The refractory-lined combustion chamber is one of the key advantages of the TKAN INTEGRA model. This design enables complete and stable fuel combustion, minimal emissions of harmful gases and dust particles, and maximum utilisation of the energy in pellets. At the same time, the boiler's steel components have no direct exposure to the flame, significantly extending boiler service life.

A cyclone reduces the quantity of particles entering the chimney. Larger configurations use a multicyclone with a centrifugal fan. The manufacturer particularly recommends a cyclone when using pneumatic heat exchanger cleaning, as this expels additional ash and soot from the boiler. Larger TKAN Integra models use a multicyclone with a centrifugal fan. Specifically, the TKAN 80 Integra and TKAN 100 Integra have a fan on the flue outlet, while the TKAN 150 Integra, TKAN 200 Integra, TKAN 250 Integra and TKAN 300 Integra have a multicyclone with a centrifugal fan.

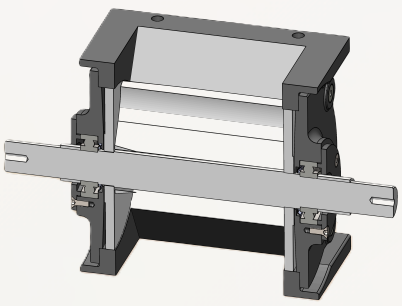
Some TKAN configurations use a fan on the flue gas side, while larger systems may have a multicyclone equipped with a centrifugal fan. This matters when designing the complete chimney system, particularly where there are several boilers, a longer flue duct, a cyclone or multicyclone, multiple bends or a shared chimney.

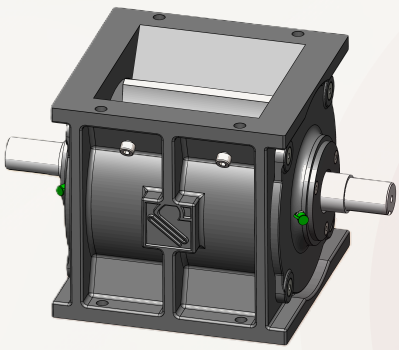


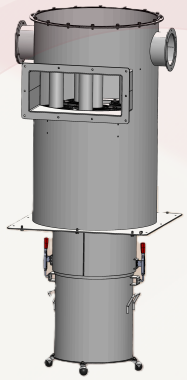
TKAN INTEGRA boilers include an automatic cleaning system for the area around the combustion chamber as standard. Compressed air automatically cleans the tubular heat exchanger. Periodic air pulses remove soot deposits from the flue tubes, maintain high boiler efficiency and significantly reduce the need for manual maintenance.

TKAN INTEGRA / TECHNICAL SPECIFICATIONS

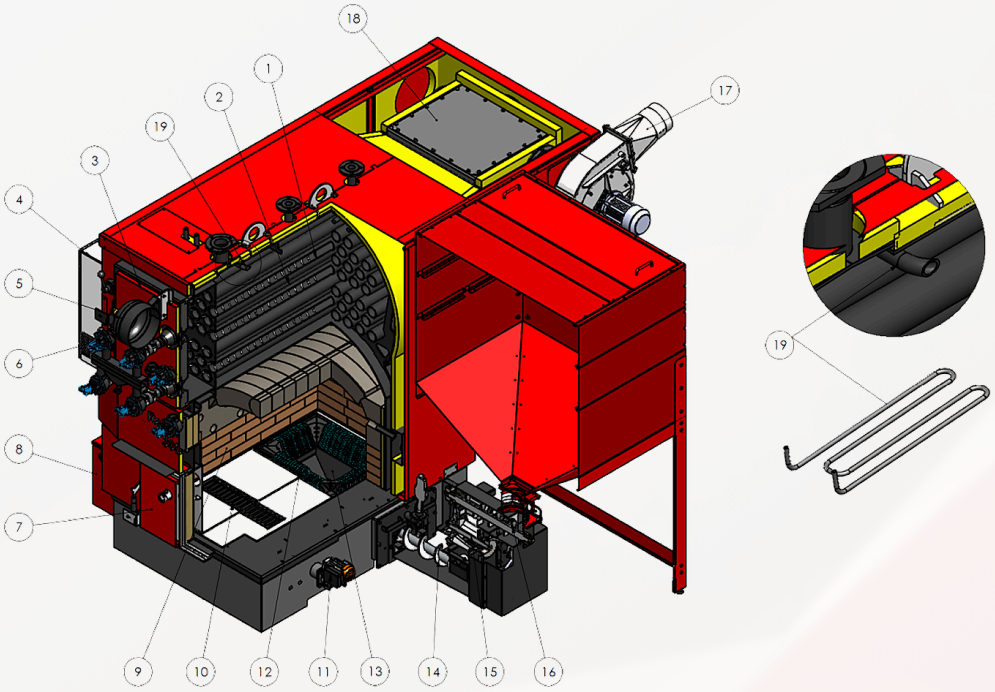
BOILER TYPE	BOILER TYPE	BOILER TYPE	TKAN 80 Integra	TKAN 100 Integra	TKAN 150 Integra	TKAN 200/250/300 Integra
		Unit				
Output	Output	kW	80	100	150	200/250/300
Operating pressure	Operating pressure	kPa	300	300	300	300
Test pressure	Test pressure	kPa	450	450	450	450
Boiler water volume	Boiler water volume	L (approx.)	368	460	690	920/1150/1380
Boiler weight	Boiler weight	kg	1191	1415	2288	3240/5293/5605
Silo weight	Silo weight	kg	100	165	165	225
	As	mm	606	1006	1006	1394
Dimensions	B	mm	1020	1456	1456	1830
Dimensions	Hs	mm	1736	1872	1872	1822
	V	L	290	680	680	1600/2330/3115
Pellet capacity of the silo	Pellet capacity of the silo	kg	180	410	410	1000/1500/2000







08 TKAN INTEGRA BOILER CROSS-SECTION



01 Turbulators

02 Tubular heat exchanger

03 Access door for cleaning the tubular heat exchanger and boiler

04 Electrical cabinet with controls

05 Compressed air vessel

06 Pulse solenoid valve

07 Fuel-loading and ignition door

08 Ash container

09 Combustion chamber refractory lining

10 Auger for automatic ash removal from the combustion chamber

11 Drive motor for the auger that automatically cleans the combustion chamber

12 Cast iron segments

13 Boiler combustion chamber

14 Lower screw conveyor shaft

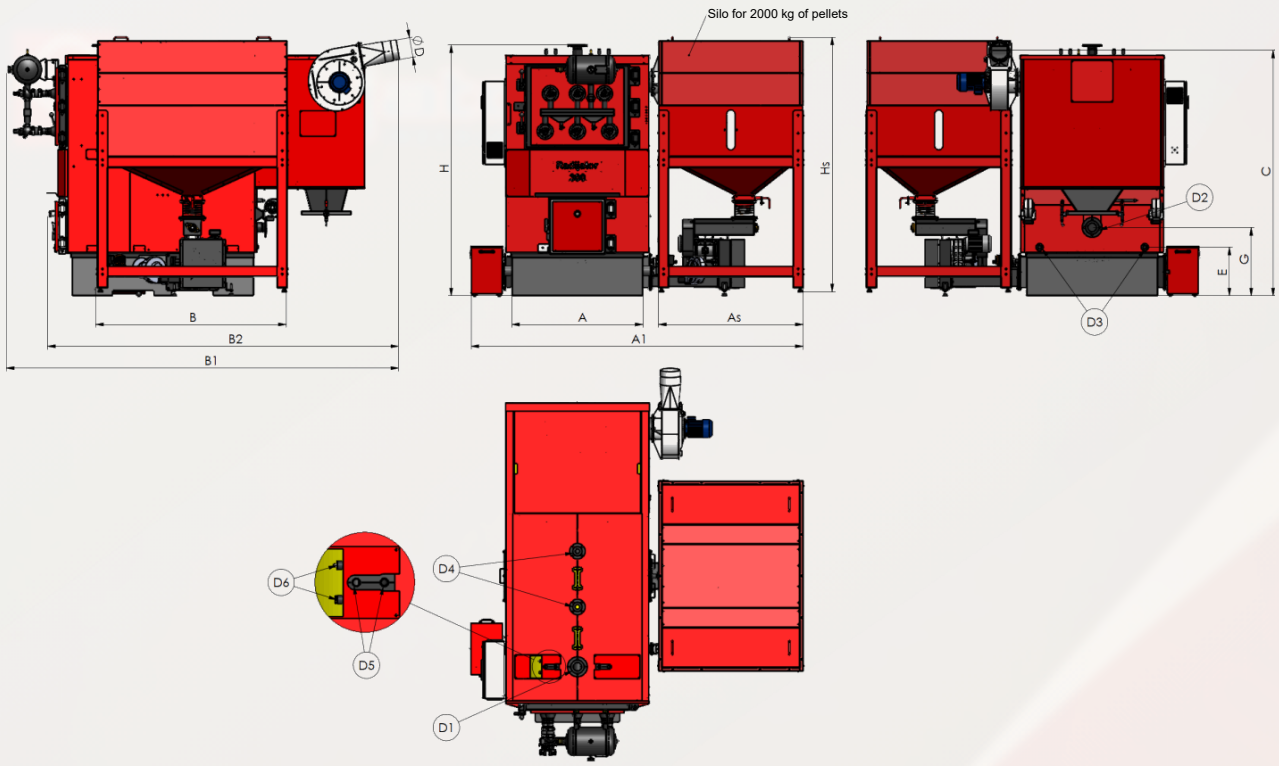
15 Rotary feeder (rotary valve)

16 Upper screw conveyor shaft

17 Multicyclone centrifugal fan

18 Multicyclone

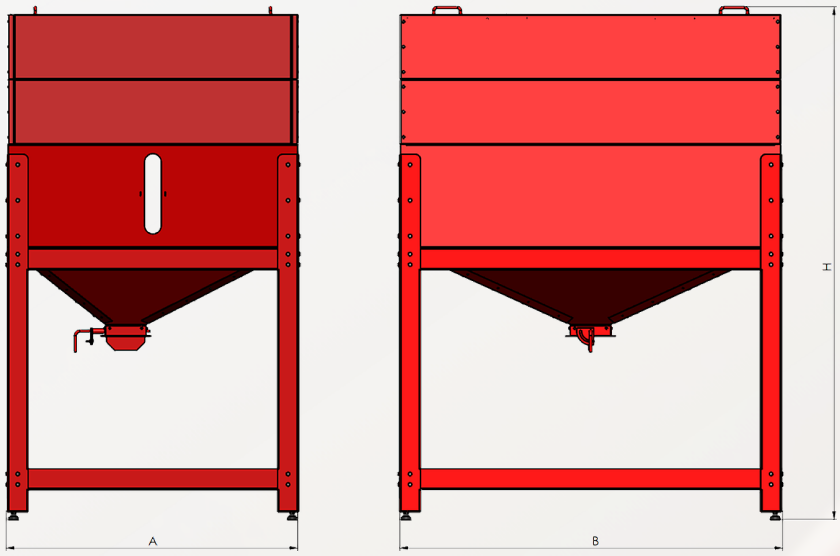
19 Thermal safety heat exchanger



TKAN INTEGRA / BOILER DIMENSIONS

DIMENSIONS	DIMENSIONS	TKAN 80 Integra	TKAN 100 Integra	TKAN 150 Integra	TKAN 200 Integra	TKAN 250 Integra	TKAN 300 Integra
A	mm	730	730	850	1005	1380	1380
A1	mm	1788	2276	2394	2942	3200	3200
As	mm	606	1006	1006	1394	1394	1394
B	mm	1020	1456	1456	1830	1830	1830
B1	mm	1655	1817	3099	3386	3755	3755
B2	mm	1595	1767	2729	2988	3360	3360
C	mm	1507	1507	2080	2515	2312	2312
ØD	mm	180	200	190	190	190	190
E	mm	390	390	359	465	467	467
G	mm	417	417	784	675	707	707
H	mm	1960	1960	2151	2547	2413	2413
Hs	in.	1736	1872	1872	1822	1822	1822
D1	in.	2"	2"	2"	DN80 NP6	DN80 NP6	DN80 NP6
D2	in.	2"	2"	2"	DN80 NP6	DN80 NP6	DN80 NP6
D3	in.	1/2"	1/2"	1/2"	DN40 NP16	DN40 NP16	DN40 NP16
D4	in.	3/4"	3/4"	3/4"	DN40 NP16	DN40 NP16	DN40 NP16
D5	in.	1/2"	1/2"	1/2"	1"	1"	1"
D6	in.	1/2"	1/2"	1/2"	1/2"	1/2"	1/2"
Mass flow rate (kg/h)	Mass flow rate (kg/h)	313	468	878	1116	1353	1461

09 TKAN INTEGRA SILO DIMENSIONS



	DIMENSIONS	DIMENSIONS	DIMENSIONS	DIMENSIONS	DIMENSIONS
	A	B	H	V	Pellet capacity of the silo
	mm	mm	mm	L	kg
TKAN 80 Integra	606	1020	1736	290	180
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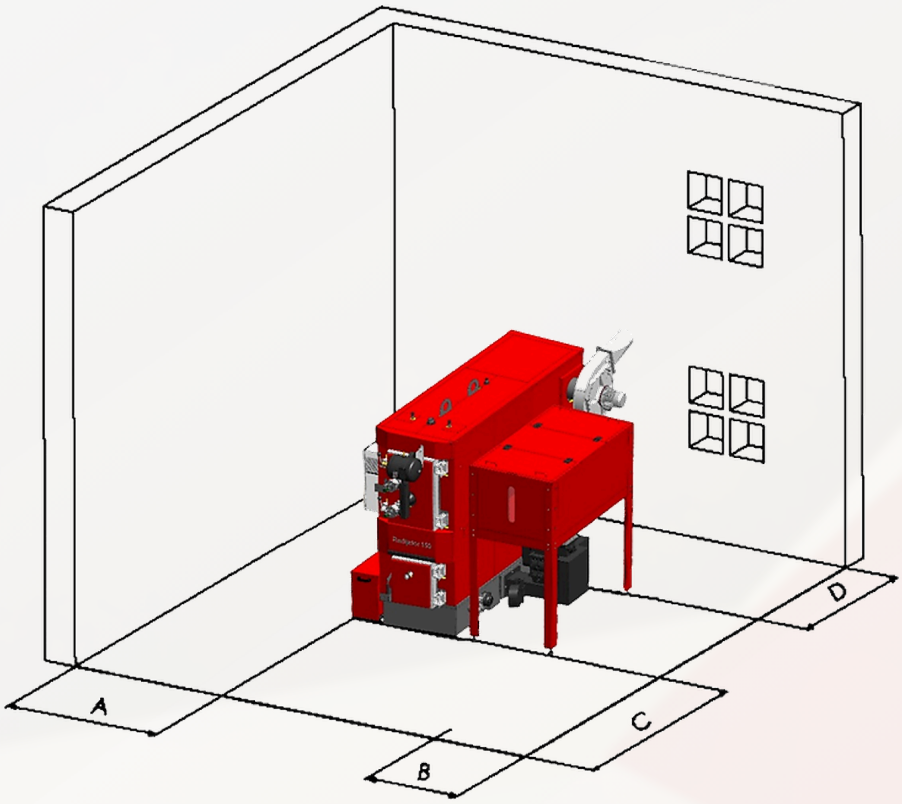
TKAN 150 Integra	1006	1456	1872	680	410
TKAN 200/250/300 Integra	1394	1830	1828	1600	1000
TKAN 200/250/300 Integra	1394	1830	2138	2330	1500
TKAN 200/250/300 Integra	1394	1830	2448	3115	2000

10 POSITIONING TKAN AND TKAN INTEGRA BOILERS IN THE BOILER ROOM

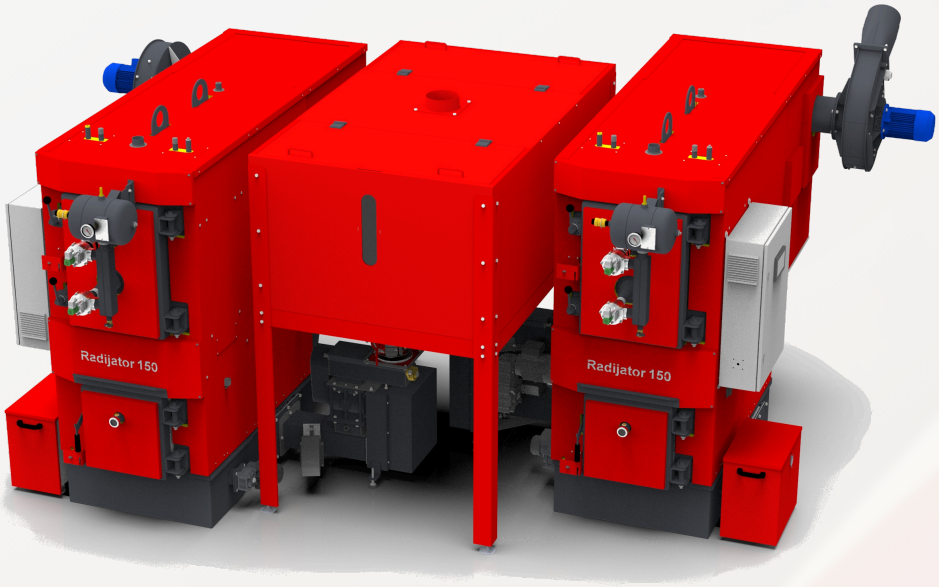


POSITIONING THE BOILER IN THE BOILER ROOM

	DIMENSIONS	DIMENSIONS	DIMENSIONS	DIMENSIONS
Boiler type	A (mm)	B (mm)	C (mm)	D (mm)
TKAN 60	100	400	1000	800
TKAN 80	500	400	1000	800
TKAN 100	500	400	1000	800
TKAN 150	500	550	1000	1000
TKAN 200	600	650	1000	1000
TKAN 250	600	900	1000	1100
TKAN 300	600	900	1000	1100
TKAN 80 Integra	500	400	1000	800
TKAN 100 Integra	500	400	1000	800
TKAN 150 Integra	500	550	1000	1000
TKAN 200 Integra	600	650	1000	1000
TKAN 250 Integra	600	900	1000	1100
TKAN 300 Integra	600	900	1000	1100

The boiler room must have protection against freezing. The boiler must stand on a non-combustible surface. The table and illustration show the recommended clearances on all four sides of the boiler from the boiler room walls or other fixed objects, such as a thermal storage tank.

11 CASCADE SYSTEMS



Radijator Inženjering developed the TKAN boiler series primarily for burning biomass, including pellets and fruit stones, and wood. Connecting these boilers in cascade systems creates a highly powerful and flexible heating solution for large buildings.

TKAN BOILERS IN CASCADE APPLICATIONS

- Cascade systems most commonly use industrial models with higher outputs, such as the TKAN 100, 150, 200, 250 and 300 kW models.
- High output requirements: Connecting two TKAN 300 boilers in a cascade, for example, creates a system with a total output of 600 kW, suitable for heating hotels, production halls or residential complexes.
- Modularity and flexibility: During autumn and spring, only one TKAN boiler operates at its optimum load, while the second starts only when the outside temperature falls significantly.

FEATURES OF TKAN CASCADE SYSTEMS

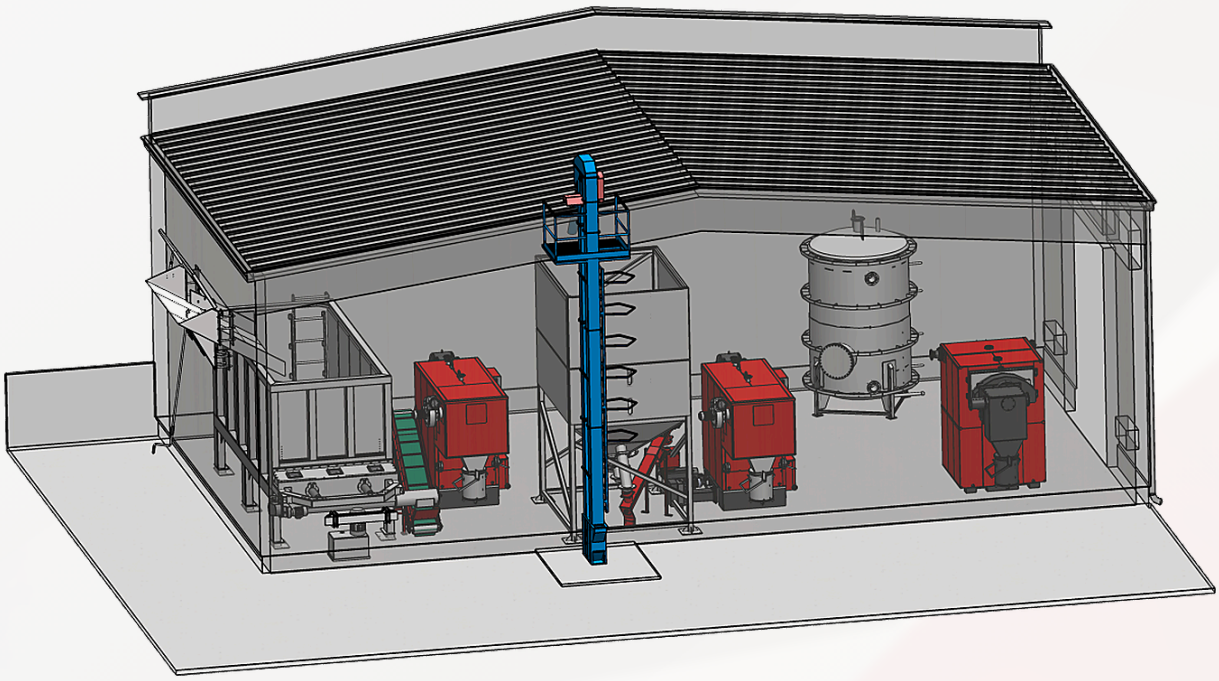
Modularity and flexibility: During autumn and spring, only one TKAN boiler operates at its optimum load, while the second starts only when the outside temperature falls significantly.

Control and automation: TKAN boilers have advanced electronics that enable synchronised operation through external cascade controllers. The controls monitor the temperature in the hydraulic separator and determine which boiler should start.

Continuous fuel supply: Industrial TKAN boilers come with day silos, for example with an 800-litre capacity, which can connect through additional screw conveyors to a single large central pellet silo supplying the entire cascade.

Reliability and continuity: If one boiler requires ash cleaning or routine servicing, the hydraulic system and cascade controls allow the other boiler to continue operating without interruption, so the building always has heating.

12 ADDITIONAL EQUIPMENT



For reliable and efficient operation of industrial boiler installations, the system can include additional components adapted to the building's requirements and operating conditions.

Additional equipment includes automatic fuel dosing and transport systems, pellet storage, automated operating controls, and equipment for connecting boilers and controlling them in cascade.

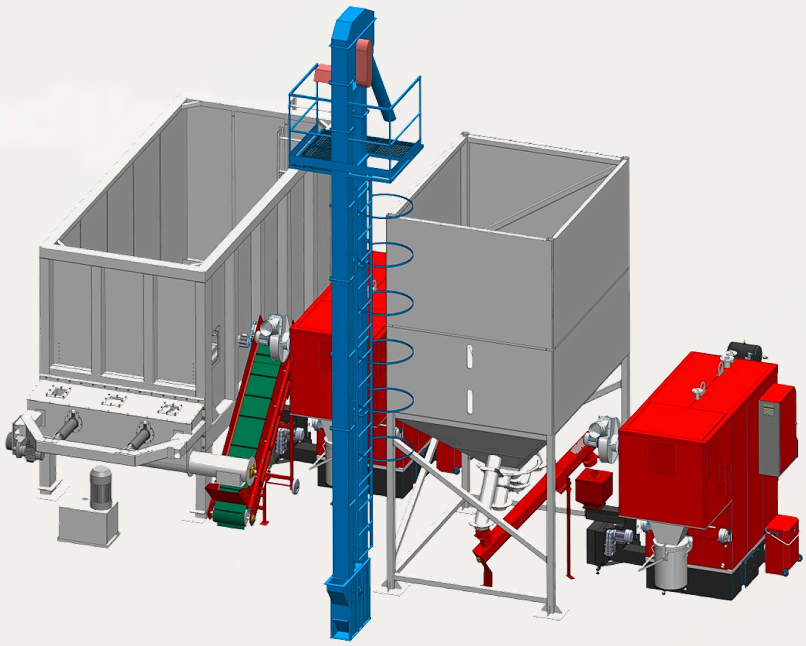
We design solutions around boiler room capacity, the required duration of unattended operation and available space, aiming for a high level of automation, reliability and optimum fuel consumption.

Two screw augers, each with its own electric drive, provide automatic ash removal in the combustion chamber area. They transfer ash into two containers that require periodic emptying.

To reduce airborne ash particle emissions, the manufacturer recommends installing a cyclone, particularly where the customer has also installed a pneumatic cleaning system.

A system of solenoid valves on the tubular heat exchanger access door periodically releases pressurised air to clean ash and soot from the boiler tubes. The system requires a compressed air supply of suitable capacity and controls to manage the process.

For large systems consuming from several hundred kilograms to several tonnes of pellets per day, the manufacturer recommends a large silo with a bucket elevator. Screw conveyors connect it to the small silo, and minimum and maximum level sensors in the small silo automate the entire delivery process.



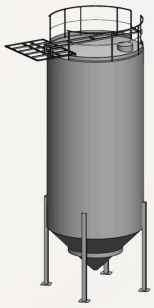
LARGER DAY SILO

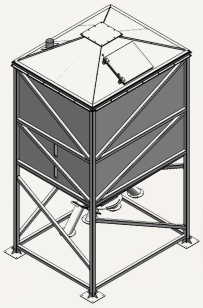
Standard TKAN boilers have a day silo. For larger systems, the manufacturer can produce larger external silos with custom dimensions. Depending on system requirements, silos with capacities of several tens of tonnes can include a bucket elevator.

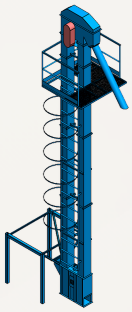
Screw conveyors connect the large external silo to the boiler's day silo, allowing automatic refilling controlled by minimum and maximum level sensors. Jumbo bags can also store larger quantities of pellets.

On newer TKAN 60-300 models, the standard day silo has a volume of 800 litres and can connect to a large external silo. The connection can run from the side or the front, depending on the equipment arrangement and available space. The fuel delivery sequence is:

Large central silo, screw conveyor, TKAN day silo, auger to the combustion chamber







13 AUTOMATIC PELLET TRANSPORT

The manufacturer provides for screw conveyors, screw conveyor drives and motors, connections between the large silo and the day silo, automatic day silo refilling, and minimum and maximum level sensors in the day silo.

For large systems, the manufacturer recommends a large silo with a bucket elevator, screw conveyors and automatic day silo refilling controlled by minimum and maximum level sensors. Jumbo bags can also store larger quantities of pellets.

The boiler controls can operate the motor of the auger conveying pellets from the large silo.

This is particularly useful for a cascade of several TKAN boilers, as it allows the design of a central pellet distribution system.

Automatic day silo refilling uses minimum and maximum level sensors. At MIN, pellet transport starts. At MAX, pellet transport stops. The boiler therefore requests fuel from central storage automatically and requires no manual refilling.

DESIGN / MANUFACTURING / SUPPORT

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